

Problem Statement

The core problem: unified, real-time oversight of three coupled risks and deciding how much verification each response is worth. Enterprises now run AI across customer support, knowledge management and agentic workflows; oversight cannot cost speed or economics.

WHY IT ISN'T SOLVED

Hallucination evals

Guardrails

Observability

Cost tools

Four tools, four answers, all after the fact, never one verdict.

THE OVERSIGHT PARADOX

Bolt on an LLM judge



Every response slower, costlier



Fly blind or pay a safety tax

Heavy verification costs more than the failures it prevents.

WHAT FAILURE COSTS

Financial

Operational

Reputational

Compliance

A support bot quoting the wrong policy, an agent looping on tool calls, an assistant leaking a record, each lands on all four.

VERIFIED 2026 EVIDENCE

01 PERFORMANCE [Confidently wrong]

- Meta, Mar 2026: an unauthorised agent's advice exposed data to the wrong engineers - Sev 1.

Proves: Damage lands when someone trusts the answer.

02 COST [Quietly expensive]

- Uber exhausted its 2026 AI budget in four months; power users ran \$2,000/month.

Proves: AI can work perfectly while quietly overspending.

03 RESPONSIBILITY [Biased, unsafe, leaking]

- Stanford, Jun 2026: across 4M applications, bias hit 26% of Black applicants.

Proves: A system can do its job and still discriminate.

WHAT'S MISSING?

Three coupled failures. One decision. From after-the-fact autopsy to watch, catch, act.

EU AI Act Art. 50 (2 Aug 2026): accountable by default.

DETECT

three risks, in real time

DECIDE

how much oversight is worth

ACT

allow, edit, escalate, block

Proposed Solution

ControlPlane.ai one verdict, three risks, any model

Oversight as a value-of-information decision under a latency budget: buy only the evidence that could change the verdict.

$\text{expected loss} = P(\text{failure}) \times \text{Cost}(\text{failure})$

HOW DO WE DETECT THE CHEAPEST SIGNAL FIRST?

	TIER 0 · free · 1–5 ms	TIER 1 · cheap · 20–60 ms	TIER 2 · costly · 200 ms → human
PERFORMANCE wrong, or overconfident?	Semantic-Entropy Probe · confidently-wrong probe	MiniCheck / HHEM groundedness vs retrieved context	N-sample self-consistency · calibrated judge
COST cheapest path to this quality?	Model-overkill · verbosity · semantic-cache hash	RouteLLM router · prompt compression	Agent-trajectory audit for loops and dead ends
RESPONSIBILITY biased, unsafe, leaking?	GLiNER-Guard safety+PII in one pass · PromptGuard-2	Opir multi-task safety · bias counterfactual	Llama-Guard-4 / ShieldGemma · human review

THE STOPPING RULE

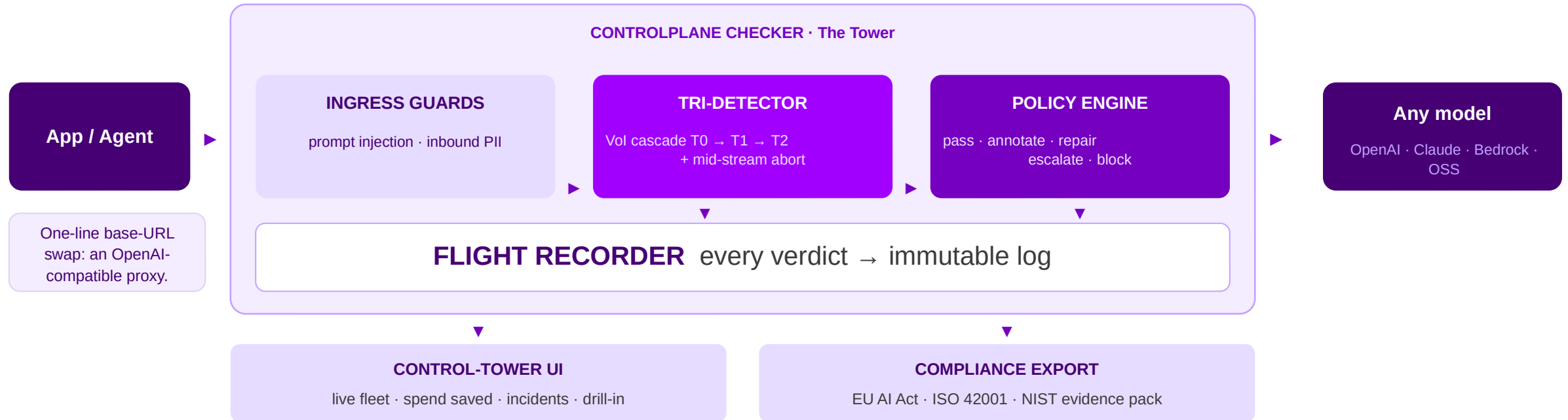
Climb a tier only when the expected loss reduction beats that check's own cost and latency.

WHY THIS DOESN'T SLOW THE AI DOWN

- Semantic-Entropy Probes cut resampling overhead to near zero — one forward pass.
- MiniCheck matches GPT-4-level fact-checking at 400× lower cost.
- RouteLLM holds 95% of GPT-4 quality routing only 14–26% to the strong model.

Solution Architecture and Action Layer

SYSTEM VIEW PROPOSED



ACTION LAYER — THE BRIEF'S SECOND QUESTION



The Cost axis isn't a third thing to watch: its savings subsidise the other two. Route-down, cache and mid-stream abort typically reclaim 30–70% of token spend, and a slice of that pays for the hallucination and safety checking. **Safer and cheaper oversight with a negative price tag. We plan to present the risk forecast and cost saved over a dashboard.**